

Credit Score Classification : Predictive Modeling for Financial Risk Assessment

Leveraging Machine Learning to Drive Informed Lending Decision

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Problem Statement & Objectives

Problem Statement

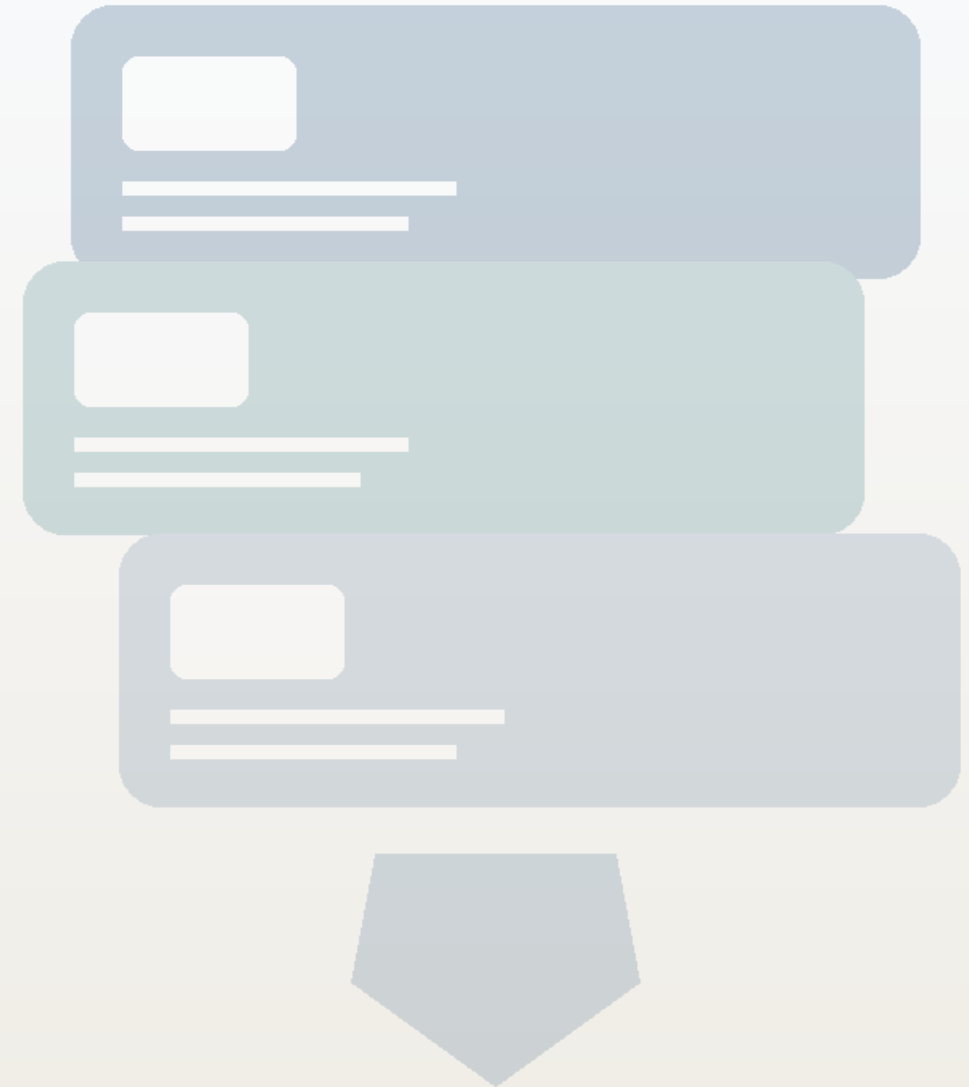
- Manual credit assessment is slow, inconsistent, and hard to scale.
- Need a data-driven model to classify customers into Poor / Standard / Good credit score categories.

Project Objectives

- Build a robust ML model using financial and personal attributes.
- Automate and standardize creditworthiness assessment.
- Improve decision speed while maintaining accuracy and fairness.

Business Impact

- Faster approvals and better customer experience.
- Consistent risk decisions across applicants.
- Improved credit risk management.



Data Overview

- Train set: 100,000 rows × 28 columns
- Test set: 50,000 rows × 27 columns
- Target: Credit_Score (Poor / Standard / Good)
- Class distribution: Standard 53% | Poor 29% | Good 18%
- Raw missing values: ~2.15% (after cleaning: ~3.69%)
- Duplicates: 0

Target Distribution



Data Quality Issues (Raw Data)

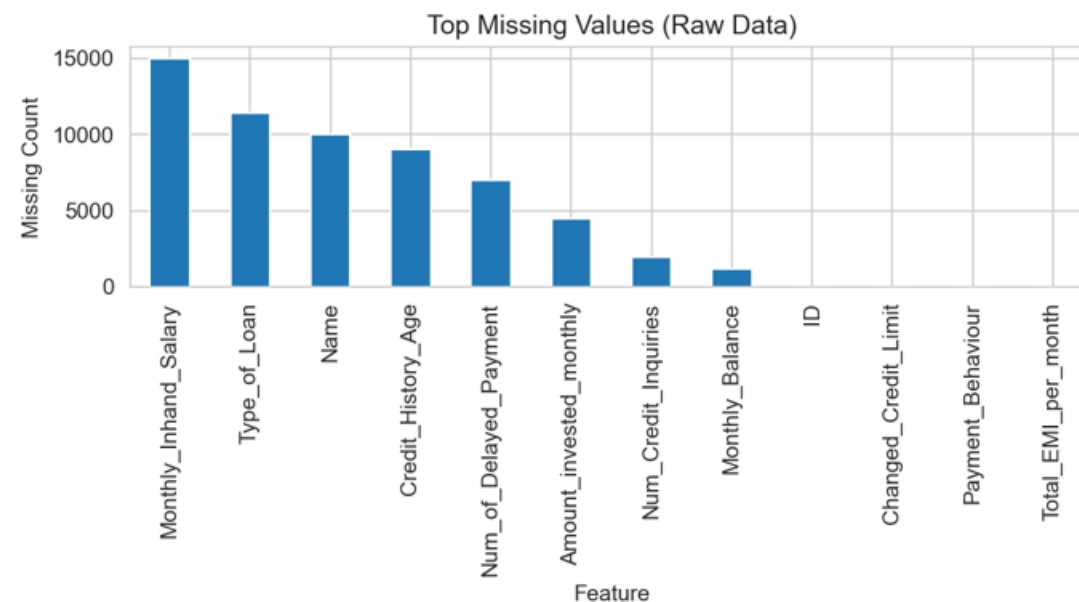
Key Issues Identified

- Mixed data types in numeric columns (e.g., Age, Annual_Income)
- Junk placeholders like '_', '_____', '!'@9#%8'
- Missing values across important features
- Multi-label text in Type_of_Loan
- Outliers in Age and loan counts

Why this matters

- The chart shows which columns have the highest missingness.
- Missing values + junk tokens reduce model reliability if not cleaned.
- Therefore, we standardize invalid tokens to NaN and impute later inside the pipeline to avoid data leakage.

Top Missing Values in Raw Data



Cleaning & Preprocessing Approach

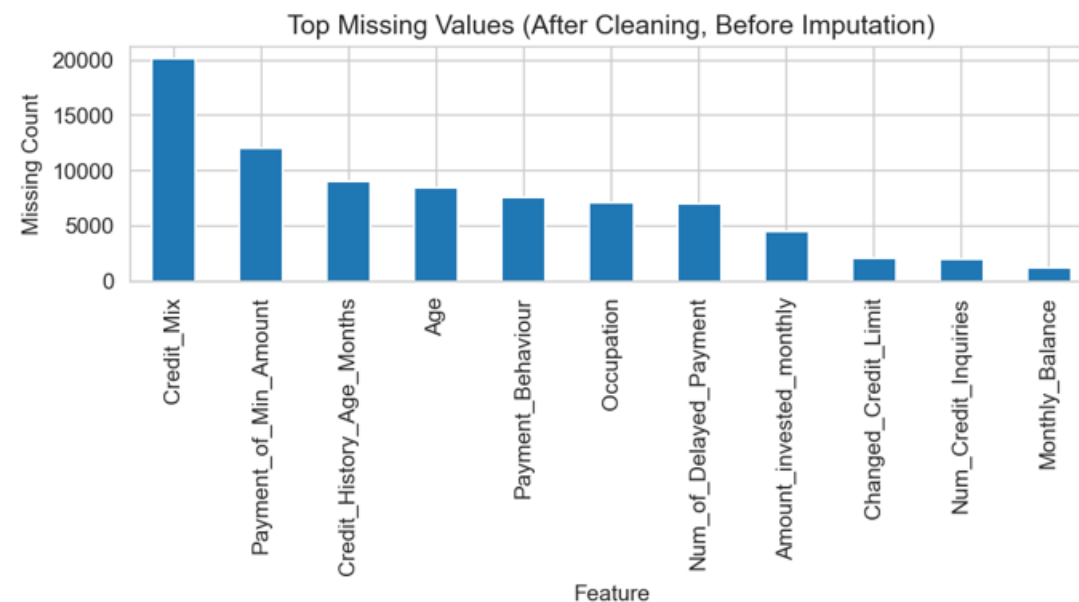
Key Cleaning Steps

- Replace junk tokens (_ , _____ , !@9#%8) with NaN
- Parse numeric fields and fix data types
- Convert Credit_History_Age to months
- Create Loan_Count from Type_of_Loan
- Drop ID/PII and redundant columns
- Keep missing values for pipeline imputation

Why this matters

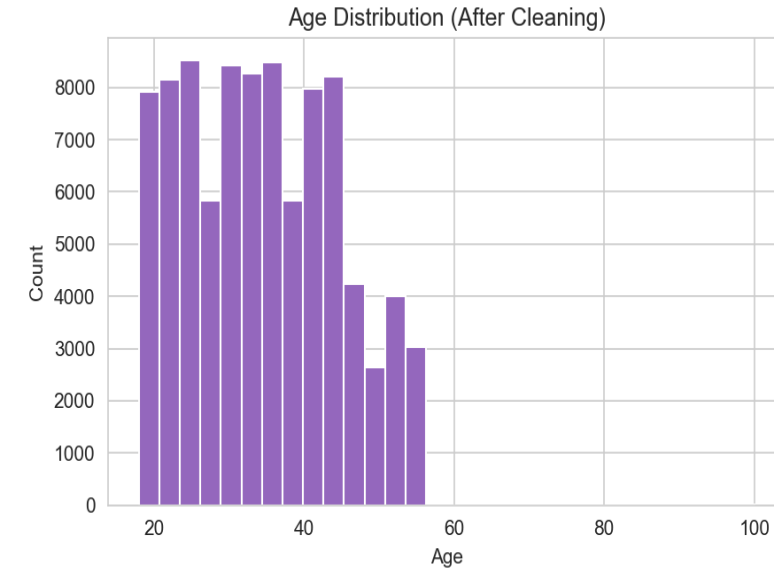
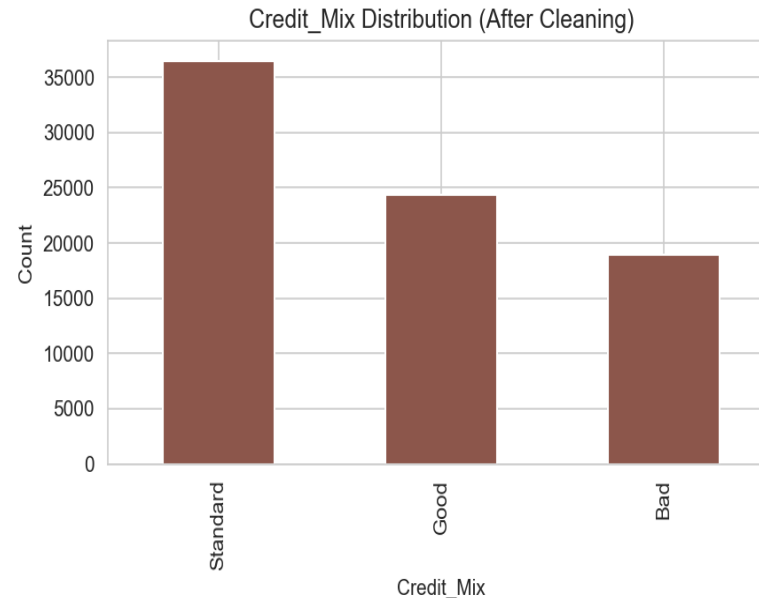
- Cleaning standardized messy inputs and removed invalid tokens.
- Missing values remain by design and are handled in the pipeline.
- This avoids data leakage and keeps all rows for modeling.

Missing Values After Cleaning (Before Imputation)





Feature Distributions (After Cleaning)



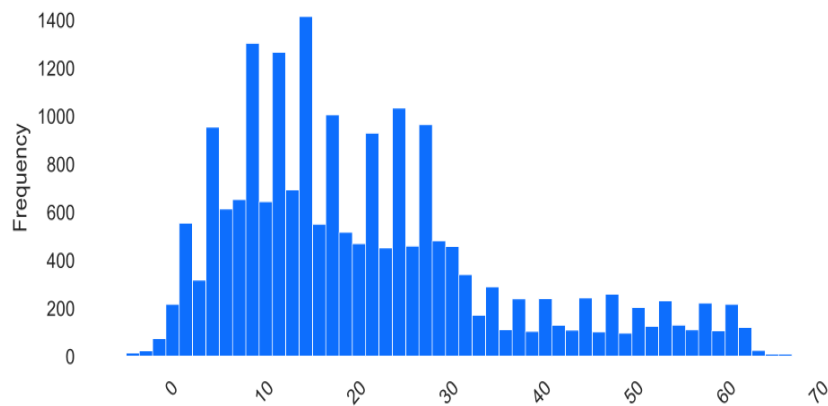
What this confirms:

- Age distribution is now realistic after removing invalid ages
- Credit Mix categories are clean and interpretable
- Payment Behaviour values are standardized (junk removed)
- Clean distributions make features reliable for modeling

Clean, consistent feature distributions improve model stability and make risk patterns easier to interpret.

Key feature signal (profiling)

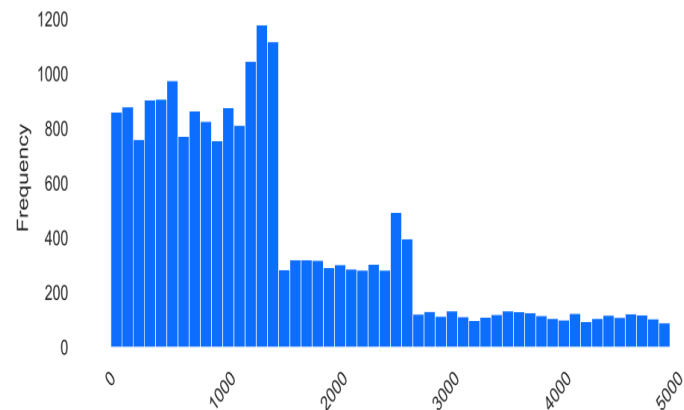
Delay from due date



Histogram with fixed size bins (bins=50)

- Shows how often and how late customers pay.
- Frequent delays reflect weaker repayment behavior.
- Useful to identify risky or borderline profiles.

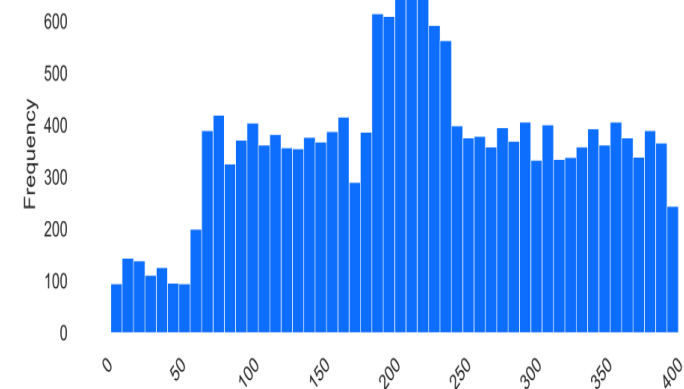
Outstanding debt



Histogram with fixed size bins (bins=50)

- Shows the current debt burden of customers.
- Higher debt generally indicates higher repayment risk.
- Helps flag customers who may need stricter review.

Credit history age



Histogram with fixed size bins (bins=50)

- Shows how long a customer has maintained credit history.
- Longer history often signals stability and predictability.
- Helps separate new vs mature credit profiles.

Model Accuracy Comparison

Key Takeaway

- Random Forest performs best (Accuracy 0.783)
- XGBoost is close behind (0.759)
- Logistic Regression is baseline (0.614)

Business Meaning

- Higher accuracy improves credit risk decisions
- Better model = fewer wrong approvals/declines
- Supports consistent, data-driven policies

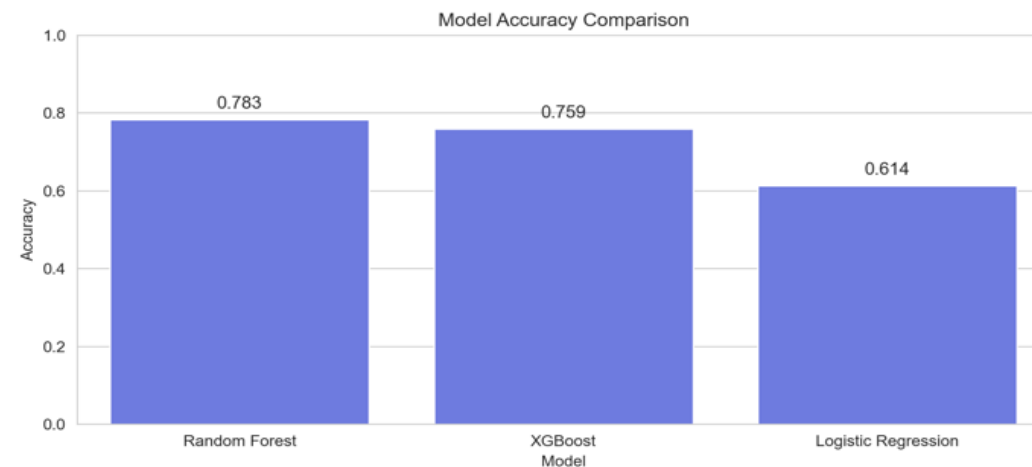
Validation Metrics

Random Forest: Acc 0.783 | BalAcc 0.766 | F1 0.768

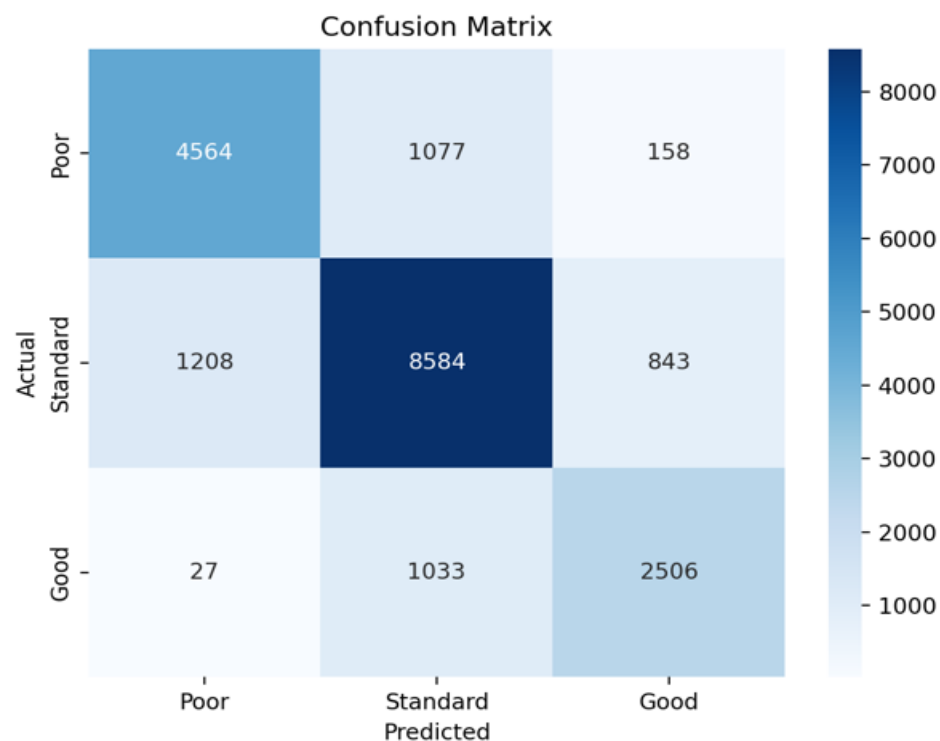
XGBoost: Acc 0.759 | BalAcc 0.722 | F1 0.724

Logistic Regression: Acc 0.614 | BalAcc 0.571 | F1 0.576

Validation Accuracy by Model



Confusion Matrix (Best Model) – What It Means



Correct Predictions (Business Meaning)

- **Poor correct: 4564** → flag high-risk customers for stricter review or lower credit limits.
- **Standard correct: 8584** → stable segment for normal approvals and pricing.
- **Good correct: 2506** → low-risk customers for faster approvals and better offers.

Common Confusions

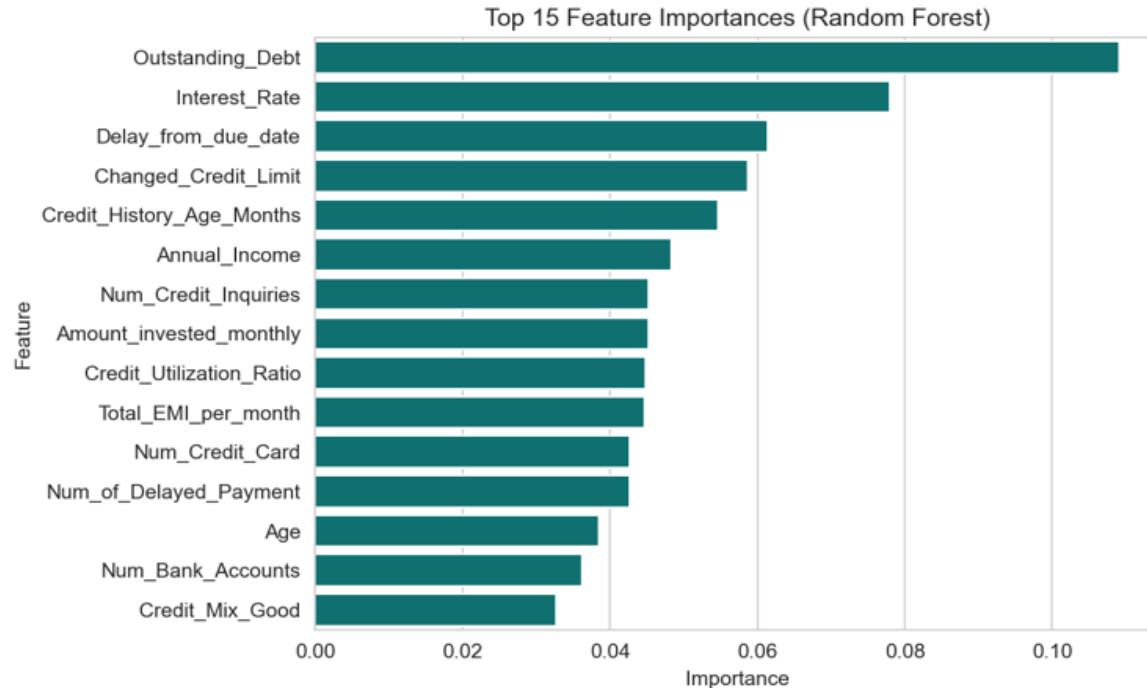
- **Poor → Standard: 1077**
- **Standard → Poor: 1208**
- **Good → Standard: 1033**

Most errors are between adjacent classes, so we can target borderline cases for review.

Why this helps

- Shows where the model is strong vs borderline
- Supports targeted manual review to reduce risk

Feature Importance (Random Forest)



Top Drivers (Importance)

- **Outstanding_Debt: 0.109**
- **Interest_Rate: 0.078**
- **Delay_from_due_date: 0.061**
- **Changed_Credit_Limit: 0.059**
- **Credit_History_Age_Months: 0.055**
- **Annual_Income: 0.048**

Business Meaning

- **Outstanding_Debt + EMI** → current repayment burden
- **Interest_Rate & Delays** → repayment risk behavior
- **Credit_History_Age** → maturity & stability signal
- **Annual_Income** → affordability indicator
- **Credit_Utilization** → reliance on available credit

How we use this

- **Focus policy checks on debt, delays, and utilization**
- **Prioritize manual review for high-burden profiles**
- **Tailor limits/terms based on affordability signals**

Key Findings & Business Impact

Key Findings (Data + Model):

- Data cleaning successfully removed junk tokens and standardized numeric fields.
- Missing values were intentionally kept and handled later via imputation to avoid data leakage.
- Random Forest performed best among all tested models.
- The model relies most on debt-related signals, payment delays, and credit history.

Business Impact (What this means):

- The model enables **faster and consistent credit decisions** compared to manual review.
- It flags **high-risk profiles** (high debt + delays), helping reduce default risk.
- It supports **transparent decision-making** by showing which factors matter most.
- This provides a strong foundation for automated credit scoring.

Business Analysis & Recommendations

-Business Problem Identified

High-risk and borderline customers are the most difficult to classify.

Misclassification here can lead to **wrong approvals or unnecessary rejections**.

-Key Business Insight

Risk signals are strongest around **debt burden, payment delays, and credit history**.

These factors explain why the model flags a customer as Poor vs Standard.

-Recommendations (Actionable)

Strengthen policy checks for customers with high debt + frequent delays.

Introduce a manual review step for borderline cases (Standard vs Poor).

Prioritize clean data capture for payment history and credit inquiries.

-Business Benefits

Lower default risk

Faster, consistent approvals

Clearer decision logic for compliance and audit

-KPIs to Track

Approval accuracy vs default rate

Misclassification rate for borderline profiles

Time saved per credit decision

Conclusion

- The project delivers a data-driven credit scoring approach that reduces manual effort.
- It highlights the most important risk factors for decision-making.
- The outcome is faster approvals, better risk control, and consistent policy execution.



Risk Gauge